

Assignment 1 – Data Import from Multiple Sources

1. What is Business Intelligence (BI)? How does data import fit into the BI lifecycle?

Business Intelligence (BI) is a process of collecting, analyzing, and visualizing data to support business decision-making.

Data import is the first step in the BI lifecycle where data is collected from multiple sources such as databases, Excel files, APIs, and cloud systems for further ETL and analysis.

2. What are the most common data sources used in a BI project? Give five examples.

Common data sources:

1. Excel files
2. CSV files
3. SQL Server databases
4. Oracle databases
5. REST APIs

3. What is a flat file? How does importing an Excel file differ from a CSV?

A flat file stores data in plain text format without relationships. Example: CSV.

Difference:

- Excel supports multiple sheets, formatting, formulas.
- CSV stores plain comma-separated text without formatting.

4. What is a database connection string? Write an example for SQL Server.

A connection string contains information required to connect to a database.

Example:

```
Server=localhost;  
Database=SalesDB;  
User Id=sa;  
Password=1234;
```

5. What is ODBC? How does it allow connection to different databases?

ODBC (Open Database Connectivity) is a standard API used to connect applications to databases using drivers.

It provides database independence through ODBC drivers.

6. What is OLEDB? How does it differ from ODBC?

OLEDB is Microsoft's technology for accessing various data sources.

Difference:

- ODBC mainly works with relational databases.
- OLEDB supports relational and non-relational data sources.

7. What is a JDBC driver? In which tools is it commonly used?

JDBC (Java Database Connectivity) driver allows Java applications to connect with databases.

Used in Java-based tools like WEKA, Talend, and Apache Spark.

8. What is Power Query in Power BI? How does it help import data from multiple sources?

Microsoft Power BI Power Query is a data transformation and import tool.

It connects to Excel, SQL, APIs, cloud storage, etc., and performs ETL operations before loading data.

9. What steps would you follow to import data from Oracle into SQL Server?

1. Install Oracle driver
2. Configure ODBC/OLEDB connection
3. Create linked server or SSIS package
4. Extract Oracle data
5. Transform data if required
6. Load into SQL Server

10. What is schema-on-read vs schema-on-write? Give an example of each.

- Schema-on-write: Structure defined before storing data. Example: SQL Server table.
- Schema-on-read: Structure applied while reading data. Example: Hadoop JSON files.

11. What is data type mismatch? How do you handle it during import?

It occurs when source and destination data types differ.
Handled using:

- Data conversion
- Casting
- Validation rules

12. What is a primary key? Why is it important when importing into a relational target?

A primary key uniquely identifies each record.
It prevents duplicate rows and maintains integrity.

13. What is data profiling? What metrics do you capture before importing?

Data profiling analyzes data quality before import.
Metrics:

- Null count
- Uniqueness
- Data types
- Duplicate records
- Range values

14. How do you import data from a REST API into Power BI?

1. Select "Get Data"
2. Choose Web/API
3. Enter API URL
4. Provide authentication
5. Transform JSON/XML data in Power Query

15. What is an SSIS package? List its main components.

SQL Server Integration Services SSIS package is a workflow used for ETL operations.
Main components:

- Control Flow
- Data Flow
- Connection Managers
- Event Handlers

16. What is a staging area in a data warehouse? Why is raw data loaded there first?

A staging area is temporary storage for raw data before transformation. It helps in cleansing, validation, and integration.

17. What is the difference between a full load and an incremental (delta) load?

- Full load: Entire dataset loaded every time.
- Incremental load: Only changed/new records loaded.

18. What is a watermark in incremental data loading? How is it implemented?

A watermark stores the last loaded timestamp or ID. New records greater than the watermark value are imported.

19. What are orphan records? How do you detect them during import?

Orphan records do not have matching parent records. Detected using LEFT JOIN or foreign key validation.

20. What is data masking? When is it applied during import?

Data masking hides sensitive information like passwords or credit card numbers. Applied during ETL before storing or sharing data.

21. What is a linked server in SQL Server? How is it used to import from Oracle?

A linked server allows SQL Server to access external databases. Oracle data can be queried directly using OPENQUERY.

22. What is the difference between structured, semi-structured, and unstructured data? Give examples.

- Structured: Fixed schema. Example: SQL tables.
- Semi-structured: Flexible schema. Example: JSON, XML.
- Unstructured: No predefined format. Example: images, videos.

23. How would you handle importing a 10GB Excel file efficiently?

- Split data into chunks
- Convert to CSV
- Use staging tables
- Use bulk import tools
- Load incrementally

24. What is data latency? How does the choice of import method affect it?

Data latency is delay between data generation and availability.
Real-time streaming gives low latency; batch import gives higher latency.

25. What is a connection manager in SSIS? What properties does it store?

A connection manager stores connection details.
Properties:

- Server name
- Database name
- Authentication
- File path

26. What is the difference between append mode and overwrite mode during import?

- Append mode: Adds new records.
- Overwrite mode: Replaces existing data.

27. How do you import JSON data into SQL Server?

Using:

- OPENJSON function
- SSIS JSON source
- Power Query transformations

28. What is a data catalog? How does it help manage multiple data sources?

A data catalog is a metadata management system that stores information about datasets.
It improves discoverability and governance.

29. What is ETL metadata? Give examples of metadata tracked during import.

ETL metadata describes ETL process information.

Examples:

- Source file name
- Load time
- Row count
- Transformation rules
- Error logs

30. What security considerations must you address when importing data from external sources?

- Authentication
 - Encryption
 - Access control
 - Data masking
 - Secure APIs
 - Malware scanning
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Assignment 2 – Data Visualization from ETL Process

1. What is data visualization in the context of a BI project? Why is it important?

Data visualization represents data using charts and dashboards for easy understanding. It helps identify trends, patterns, and business insights quickly.

2. What is the difference between a report and a dashboard in Power BI?

- Report: Detailed multi-page analysis.
- Dashboard: Single-page summary with KPIs.

3. What is a KPI? Give three examples used in sales analytics.

KPI (Key Performance Indicator) measures business performance.

Examples:

- Total Sales
- Profit Margin
- Customer Retention Rate

4. What chart type is best for showing trends over time? Why?

Line chart because it clearly shows continuous changes over time.

5. When would you use a bar chart vs a column chart?

- Bar chart: Compare categories with long labels.
- Column chart: Compare values across time periods.

6. What is a heatmap? How is it useful after the ETL process?

A heatmap uses colors to show intensity or patterns in data.
Useful for identifying high and low performance areas.

7. What is a scatter plot? What relationship does it help visualise?

Scatter plot shows relationship between two numerical variables.
Used to identify correlation.

8. What is a pie chart vs a donut chart? When should you avoid using a pie chart?

- Pie chart: Circular chart showing proportions.
- Donut chart: Pie chart with center space for extra information.
Avoid pie charts when categories are too many.

9. What is DAX in Power BI? Give an example of a calculated measure.

DAX (Data Analysis Expressions) is a formula language in Power BI.

Example:

```
Total Sales = SUM(Sales[Amount])
```

10. What is Power Query M language? How is it used after ETL?

M language is used in Power Query for data transformation and cleansing.

11. What is drill-through in Power BI? How does it help analysts?

Drill-through allows navigation from summary data to detailed records.

12. What is a slicer in Power BI? How does it interact with visuals?

A slicer is a filtering visual that dynamically filters dashboard visuals.

13. What is row-level security (RLS) in Power BI? Why is it important after ETL?

RLS restricts users to viewing only authorized data.

14. What is a calculated column vs a measure in Power BI? When to use each?

- Calculated Column: Computed row-by-row.
- Measure: Dynamic aggregated calculation.

15. What is a star schema? How does it improve visualisation performance in Power BI?

A star schema contains one fact table connected to multiple dimension tables. It improves query speed and simplifies relationships.

16. What is data granularity? How does the grain of your ETL output affect visualisations?

Granularity is the level of detail in data. Fine-grained data gives detailed visuals but increases complexity.

17. What is a waterfall chart? Give a business use case.

Waterfall chart shows cumulative effect of positive and negative values. Use case: Profit analysis.

18. What is a funnel chart? Where is it typically used in BI dashboards?

Funnel chart shows stages in a process. Used in sales pipelines.

19. What are bookmarks in Power BI? How do they improve dashboards?

Bookmarks save dashboard states and improve navigation.

20. What is conditional formatting in Power BI tables? Give an example.

Conditional formatting changes appearance based on values.

Example: Highlight sales below target in red.

21. What is the difference between a live connection and an import mode in Power BI?

- Live connection: Real-time data access.
- Import mode: Data stored inside Power BI.

22. What is a decomposition tree visual? How does it help with root cause analysis?

It breaks down measures into contributing factors interactively.

23. What is the importance of colour theory in data visualisation?

Proper colors improve readability and reduce confusion.

24. What is a sparkline? Where are they useful in dashboards?

A sparkline is a mini chart inside a table cell showing trends.

25. How do you publish a Power BI report to the Power BI Service?

Use the Publish button from Power BI Desktop.

26. What is a paginated report in Power BI? How does it differ from standard reports?

Paginated reports are printable pixel-perfect reports.

27. How do you schedule automatic data refresh in Power BI Service?

Configure gateway and set refresh schedule in Power BI Service.

28. What is the Q&A visual in Power BI? Give an example query.

Q&A visual allows natural language queries.

Example: "Show total sales by region."

29. What are tooltips in Power BI? How can you customise them?

Tooltips display extra information on hover.
Custom pages can be used as tooltips.

30. How does data accuracy in the ETL process directly impact visualisation quality?

Incorrect ETL data leads to misleading dashboards and wrong business decisions.

Assignment 3 – ETL Process

1. What is ETL? Explain each phase with an example.

ETL stands for Extract, Transform, Load.

- Extract: Collect data from sources.
- Transform: Clean and process data.
- Load: Store data into warehouse.

Example: Extract sales data → remove duplicates → load into warehouse.

2. What is a data warehouse? How does it differ from an OLTP database?

A data warehouse stores historical analytical data.
OLTP handles daily transactions.

3. What is SSIS? What are its main components?

SSIS is Microsoft's ETL tool.

Components:

- Control Flow
- Data Flow
- Connection Managers

4. What is the difference between OLTP and OLAP?

- OLTP: Transaction processing.
- OLAP: Analytical querying.

5. What is a Data Flow Task in SSIS? What transformations can it contain?

It handles movement and transformation of data.

Transformations:

- Lookup
- Sort
- Derived Column
- Aggregate

6. What is a Lookup Transformation in SSIS? Give a real-world use case.

Lookup matches input data with reference data.

Example: Match customer IDs with customer table.

7. What is a Derived Column transformation in SSIS? Give an example.

Creates new columns using expressions.

Example: FullName = FirstName + LastName.

8. What is a Slowly Changing Dimension (SCD)? Explain Type 1 and Type 2.

SCD manages historical dimension data.

- Type 1: Overwrites old data.
- Type 2: Preserves history using new rows.

9. What is a surrogate key? Why is it preferred over a natural key?

A surrogate key is a system-generated key.

Preferred because it is stable and independent of business changes.

10. What is a fact table? What is a dimension table? Give examples.

- Fact table: Stores measurable data. Example: Sales.
- Dimension table: Stores descriptive data. Example: Customer.

11. What is the difference between a star schema and a snowflake schema?

- Star schema: Denormalized dimensions.
- Snowflake schema: Normalized dimensions.

12. What is a Control Flow in SSIS? Name three control flow tasks.

Control Flow manages workflow execution.

Tasks:

- Execute SQL Task
- File System Task
- Data Flow Task

13. What is an Execute SQL Task in SSIS? Give two use cases.

Runs SQL queries during ETL.

Use cases:

- Truncate table
- Update records

14. What is incremental loading in ETL? How do you implement it using a watermark?

Only changed data is loaded using last processed timestamp.

15. What is data cleansing? Name five common cleansing operations.

Improving data quality.

Operations:

- Remove duplicates
- Handle nulls
- Standardize formats
- Correct spelling
- Remove invalid data

16. What is a null value? How do you handle nulls during transformation?

Null means missing data.

Handled using default values, replacement, or filtering.

17. What is deduplication in ETL? How do you detect and remove duplicate records?

Removing repeated records using DISTINCT, grouping, or matching keys.

18. What is data normalisation vs denormalisation in ETL?

- Normalisation: Reduce redundancy.
- Denormalisation: Improve query performance.

19. What is an error row redirect in SSIS Data Flow?

Invalid rows are redirected to error tables instead of failing the package.

20. What is the difference between truncate-and-reload vs upsert?

- Truncate-and-reload: Delete all and reload.
- Upsert: Update existing and insert new records.

21. Write a SQL MERGE statement example.

```
MERGE TargetTable AS T
USING SourceTable AS S
ON T.ID = S.ID
WHEN MATCHED THEN
UPDATE SET T.Name = S.Name
WHEN NOT MATCHED THEN
INSERT (ID, Name)
VALUES (S.ID, S.Name);
```

22. What is package configuration in SSIS? Why is it used?

Stores configurable values externally for portability and security.

23. What is logging in SSIS? What events do you typically log?

Logging records package execution details.

Events:

- Errors
- Warnings
- Start time
- End time

24. What is Power BI Dataflow? How does it relate to ETL?

Dataflow is a cloud ETL service in Power BI.

25. What is a staging database? How does it differ from a data warehouse?

Staging database temporarily stores raw data; warehouse stores processed analytical data.

26. What is partitioning in a data warehouse table? How does it improve ETL performance?

Partitioning divides tables into smaller parts improving query and load performance.

27. What is a checksum in ETL? How is it used?

Checksum generates hash values to detect changed records.

28. What is the difference between Type 3 SCD and Type 2 SCD?

- Type 2: Full history maintained.
- Type 3: Limited history using additional columns.

29. What is a business key vs a surrogate key?

- Business key: Real-world unique identifier.
- Surrogate key: System-generated identifier.

30. What is data lineage? Why is it important in BI?

Data lineage tracks movement of data from source to destination for auditing and debugging.